

Bulk RNA-seq analysis report

24 samples sequenced, 23 analysed · GRCh38, GENCODE v44 · DESeq2 with batch in the design · quality control, model, results, sensitivity analysis and the methods section

This is a redacted sample. It shows the structure and level of detail of a real deliverable. Client, organism and study details have been removed or changed. Ask us and we will send one built around an analysis like yours.

1 What was run, and against what

Everything below was fixed in the analysis plan before the sample groups were unblinded. The plan, this report and the code produce the same numbers on a rerun; that is the point of recording them at this level rather than summarising them.

Element	Value
Design	12 treated against 12 control, single tissue, one timepoint
Reference	GRCh38 primary assembly, GENCODE release 44 — 62,700 annotated gene features
Alignment	STAR 2.7.11a, two-pass, default splice handling
Quantification	featureCounts, gene level, reverse-stranded
Model	DESeq2 1.42.0, design ~ batch + condition
Thresholds	Adjusted p < 0.05 (Benjamini–Hochberg) and log ₂ fold change > 1
Environment	Nextflow pipeline, Singularity containers, seeds fixed, lock file shipped

The reference build and the annotation release are given because they change the answer. The same reads quantified against GENCODE v38 and v44 do not produce the same gene counts, and a reviewer who cannot reproduce a count with the information in the methods will assume the less charitable explanation.

2 Quality control, and the one exclusion

Metric	Result	Note
Read pairs per sample	mean 38.2M range 31.4–44.7M	All above the 25M planned minimum.
Retained after trimming	97.1%	Adapter content negligible across the run.
Uniquely mapped	mean 91.6% range 87.3–94.1%	Excluding the sample below, which mapped at 74.8%.
Duplication	mean 22.4%	Consistent with library complexity at this input.
Genes passing the filter	14,382 of 62,700	Filter applied before groups were compared.

Metric	Result	Note
Excluded	1 sample	RIN 5.2 against a cohort median of 8.4, with marked 3' bias and 41% exonic reads against a cohort mean of 78%. Excluded on these metrics, recorded before the group label was looked at.

The exclusion rule was written down first, and it matters which way round that happened. Dropping a sample after seeing which arm it belongs to is not a quality control decision, and a reviewer cannot tell the difference from the finished table — which is why the rule, the metrics and the date are all in the report. The sample was in the control arm. Section 5 reports what the result looks like if it is kept.

Analysed set: 23 samples, 12 treated and 11 control.

3 The batch effect, and what was done about it

Principal component analysis was inspected before any model was fitted. The first component did not separate the conditions.

Observation	Value
PC1	24.1% of variance — corresponds to library preparation batch, not condition
PC2	11.3% of variance — corresponds to condition
Batch structure	Two preparation batches, 12 and 11 samples, partially crossed with condition
Handling	Batch included as a term in the design, rather than corrected before modelling
Genes called with batch in the design	699
Genes called without it	158

This is the single most consequential decision in the analysis. Omitting the batch term would have produced 158 genes and a publishable-looking result. Including it produces 699 and a correct one. Because the batches are only partially crossed with condition, some of the condition effect is not separable from batch; that limitation is stated in the manuscript rather than left implicit, and the design that would have avoided it is noted for the next study.

4 Results

Result	Value
Genes tested	14,382
Differentially expressed	699 (adjusted $p < 0.05$ and $ \log_2FC > 1$)
Higher in treated	412
Lower in treated	287
Fold changes reported	Shrunken (apeglm) for ranking; unshrunk values in the full table
Enrichment background	The 14,382 genes tested — not the annotated genome

The background matters more than it looks. Run against all 62,700 annotated features, this gene list returns a long and impressive list of enriched terms, most of which describe what is abundantly expressed in the tissue rather than what changed. Against the tested background the list is shorter, less striking and correct. This is the commonest analysis error in published transcriptomics.

5 Sensitivity analyses

Each analysis below was specified in the plan. All of them are reported, including the one that is least flattering.

- **Keeping the excluded sample.** 512 genes are called, of which 451 are in the main set of 699. The direction of effect is unchanged for all 451. The exclusion therefore increases power and does not create the finding — which is the question a reviewer is actually asking.
- **An alternative quantifier.** Salmon in place of STAR and featureCounts gives 681 genes, with 94% agreement on the called set. The conclusion does not depend on the quantification route.
- **An alternative filter.** Raising the expression filter reduces the tested set to 12,906 and the called set to 654, with no change in the enriched pathways.
- **Leave-one-out.** No single sample accounts for more than 4% of the called genes, so the result is not carried by one outlier.

6 What is handed over

- The raw and normalised count matrices, and the full results table for all 14,382 genes tested rather than the 699 that were called.
- The Nextflow pipeline, the container definitions, the lock file and the fixed seeds — enough to reproduce every number in this report.
- Per-sample quality control, including the metrics behind the exclusion and the date they were recorded.
- Publication-ready figures in vector format: PCA, volcano, heatmap and enrichment, each with the script that produced it, in a palette that survives greyscale and common colour blindness.
- A methods section written to journal length, with versions, parameters, thresholds and the accession already in it.
- The submission package for GEO, with MINSEQE-level metadata and the processed matrix alongside the raw reads.

Pubrica runs the analysis; the biological interpretation and the authorship remain yours. Where a bioinformatician's contribution meets ICMJE authorship criteria we say so and let you decide. Any AI assistance in code or drafting is declared.

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